

Name: Key

/ 25

Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

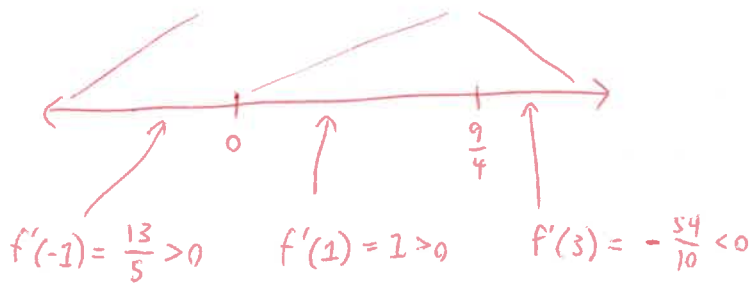
1. [11 points] Let $f(x) = -\frac{1}{10}(2x^2)(x^2 - 3x)$. Note that we have the following:

$$f'(x) = -\frac{1}{10}(2x^2)(4x - 9) \quad \text{and} \quad f''(x) = -\frac{1}{10}(12x)(2x - 3)$$

You must show your work for all parts to receive credit!

- a. Determine the intervals where f is increasing and where f is decreasing.

Critical points: $f'(x) = 0 \Rightarrow -\frac{1}{10}(2x^2)(4x - 9) = 0 \Rightarrow x = 0 \text{ or } x = \frac{9}{4}$



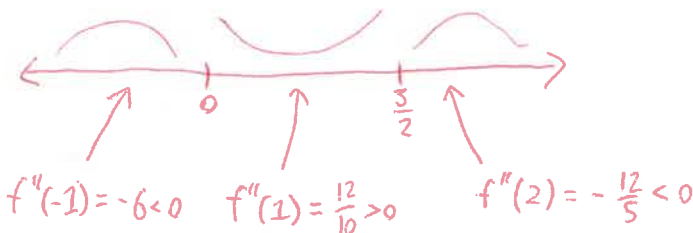
Increasing: $(-\infty, 0) \cup (0, \frac{9}{4})$
 Decreasing: $(\frac{9}{4}, \infty)$

- b. Find the x -values where any local maxima occur and where any local minima occur. If none exist, state "none".

Local max at $x = \frac{9}{4}$, No local minima (Justification above)

- c. Find the intervals where f is concave up and where f is concave down.

$$f''(x) = 0 \Rightarrow -\frac{1}{10}(12x)(2x - 3) = 0 \Rightarrow x = 0 \text{ or } x = \frac{3}{2}$$



Concave up: $(0, \frac{3}{2})$
 Concave down: $(-\infty, 0) \cup (\frac{3}{2}, \infty)$

- d. Find the x -values where any inflection points occur. If none exist, state "none".

Inflection points at $x = 0$ and $x = \frac{3}{2}$

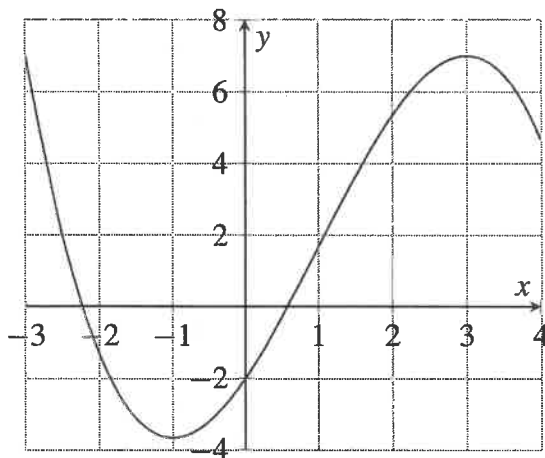
2. [8 points] Evaluate the following limits. Show your work!

$$\text{a. } \lim_{x \rightarrow -\infty} \frac{5x^3 - 7x^5}{8x^5 + 10x^4} = \lim_{x \rightarrow -\infty} \frac{\frac{5}{x^2} - 7}{8 + \frac{10}{x}} = \boxed{-\frac{7}{8}}$$

$$\begin{aligned} \text{b. } \lim_{x \rightarrow -\infty} \frac{3(x^2 + 8x)}{\sqrt{x^4 + 2x^2}} &= \lim_{x \rightarrow -\infty} \frac{3x^2 + 24x}{\sqrt{x^4(1 + \frac{2}{x^2})}} = \lim_{x \rightarrow -\infty} \frac{x^2(3 + \frac{24}{x})}{x^2 \sqrt{1 + \frac{2}{x^2}}} = \lim_{x \rightarrow -\infty} \frac{3 + \frac{24}{x}}{\sqrt{1 + \frac{2}{x^2}}} \\ &= \frac{3}{\sqrt{1}} = \boxed{3} \end{aligned}$$

Note that for negative x -values, $\sqrt{x^4} = \sqrt{x^2} \sqrt{x^2} = (-x)(-x) = x^2$

3. [6 points] Based on the graph of the function $g(x)$ below, indicated whether each of the given values is **positive**, **negative**, **zero**, or **undefined**.



a. $g'(-1) = 0$

d. $g''(-1)$ positive

b. $g'(0.5)$ positive

e. $g''(0.5) = 0$

c. $g'(3) = 0$

f. $g''(3)$ negative